7.1 Answers: numbers to a million

← Unit 7.1 · Practice

When an answer is different, look at *how* he worked before saying anything. Often the method is nearly right and one small step slipped.

Set 1 · Warm-up: big numbers

1. 10 000 (one more than 9 999 makes the next ten-thousand)
2. 25 000
3. 100 000
4. 60 000 (the 6 is in the ten-thousands column)
5. 80 000 (the 8 is in the ten-thousands column; the 4 is worth 400 000)
6. 306 200 (306 thousands, then 200; the zero keeps the tens of thousands empty)
7. 250 500 (249 thousand + 1 thousand = 250 thousand)
8. 493 000 (503 thousand − 10 thousand = 493 thousand)
9. 54 000 · 405 000 · 450 000 · 500 400 (54 000 has only five digits, so it is smallest)
10. 1 000 000

Set 2 · Warm-up: the number line

1. 500 000
2. 250 000
3. 45 000
4. 450 000 · 550 000 · 650 000
5. 175 000 · 150 000 · 125 000
6. 100 000 (1 000 000 : 10)
7. 380 000 and 390 000
8. 390 000 (386 000 is 6 000 past 380 000 but only 4 000 from 390 000)
9. 1 000 000

Set 3 · Building: rounding

1. 4 000 (the hundreds digit is 3, round down)
2. 5 000 (hundreds digit 7, round up)
3. 5 000 (exactly halfway, and halfway rounds up)
4. 38 000 (hundreds digit 2, round down)
5. 40 000 (thousands digit 8, round up)
6. 61 000 (only the hundreds digit, 4, decides; it is below halfway)
7. 300 000 (ten-thousands digit 5, round up)
8. 250 000 (thousands digit 4, round down)
9. 55 000 (hundreds digit 8, round up)
10. € 1 200 (tens digit 4, round down)
11. 1 000 (900 is closer to 1 000 than to 0)

Talk about it: both are right. It depends what you round to. Rounding to ten-thousands is rougher, so the answer is further from the real number.

Set 4 · Building: × and : by 10, 100 and 1000

1. 350 (digits move one place left)
2. 3 500 (two places left)
3. 35 000 (three places left)
4. 480 (one place right)
5. 48 (two places right)

6. 72 (three places right)
7. 60 000
8. 25 000
9. 90 (9 000 : 100 = 90)
10. 500
11. 45 crates (4 500 : 100)
12. 70

Set 5 · Stretch: word problems

1. 96 400 (86 400 + 10 000: only the ten-thousands digit changes)
2. 100 150 km (99 850 + 150 = 100 000, then 150 more)
3. € 100 000 (1 000 000 : 10)
4. 1 000 times (1 000 × 1 000 = 1 000 000)
5. a. 975 310 (biggest digits first) b. 103 579 (smallest non-zero digit first, then the 0, then the rest in order)
6. 20 000 (thousands digit 4, round down)
7. About 250 000 passengers (250 × 1 000)
8. True (450 000 is exactly halfway, and halfway rounds up)
9. Half a million (500 000 is more than 400 000)

Puzzle corner

- **A million seconds:** about 11,6 days, so just over a week and a half (1 000 000 : 86 400 seconds in a day). Most people guess much too high or much too low; both are interesting.
- **A million sheets:** 1 000 000 × 0,1 mm = 100 000 mm = 100 metres. That is taller than most church towers in the Netherlands.
- **Make your own:** any answer is fine. (A million cents is € 10 000, so that one is a good deal.)